COURSE: **CHE 312: Transport Phenomena II**

Spring Semester 2020

Department of Chemical Engineering

University of Illinois at Chicago

Chicago, Illinois 60607-7052

Homework Assignment #1

January 13, 2020

(due January 22, 2020)

Do all problems shown below.

The differential equations given below will arise frequently in various problems in this course. You should be able to solve all of them. In these equations, the b's are constants and f, g, P, and Q are arbitrary functions.

a. $y' = f(x)g(y)$

b. $y' + P(x)y = Q(x)$

c. $y'' + b^2y = 0$

d. $y'' - b^2y = 0$

e. $y'' = b_0 + b_1x + b_2x^2$

f. $\frac{1}{x}\frac{d}{dx}(x\frac{dy}{dx}) = b_0 + b_1x + b_2x^2$

g* $\frac{1}{x^2}\frac{d}{dx}(x^2\frac{dy}{dx}) + b^2y = 0$

h* $\frac{1}{x^2}\frac{d}{dx}(x^2\frac{dy}{dx}) - b^2y = 0$

i. $y'' + b_1y' + b_2y = 0$

j. $y'' + 2xy' = 0$

k. $\frac{d}{dx}[f(y)\frac{dy}{dx}] = 0$

l. $\frac{1}{x^2}\frac{d}{dx}(x^2\frac{dy}{dx}) = b_0 + b_1x + b_2x^2$

* Try the substitution $y(x) = u(x)/x$